

ASSIGNMENT 1

Cloud System Design & Pricing: Architecting a Data Platform on AWS

Assignment Overview

In this individual assignment you will assume the role of a cloud solutions architect. You will select a realistic business domain, design a complete AWS-based data platform to serve that domain, justify every architectural choice with concrete technical reasoning, and produce a defensible cloud cost estimate. The output is a presentation slide deck (10–15 slides) that could be handed to an engineering team to begin implementation.

[IMPORTANT] This assignment sets the architectural foundation for Assignments 2 and 3. The domain you choose here will carry through the rest of the course. Pick something you are genuinely interested in — your cloud architecture, data pipeline, and analytics will all be grounded in this domain.

Learning Objectives

- **Problem Framing:** Identify and articulate a data-intensive business problem and its functional requirements.
- **Solution Architecture:** Translate business requirements into a cloud architecture using appropriate AWS services (databases, storage, compute, analytics, ML).
- **Data Source Analysis:** Identify and categorize relevant data sources (transactional, streaming, batch, third-party/API).
- **Capacity Planning:** Estimate data volumes, storage needs, and compute capacity requirements based on realistic business assumptions.
- **Cloud Cost Estimation:** Generate an itemized AWS cost estimate using the AWS Pricing Calculator and reason about cost reduction strategies.
- **Technical Communication:** Communicate architectural decisions clearly and persuasively to a mixed technical/business audience.

Step 0: Choose Your Domain (Do This First)

Select ONE domain from the list below, or propose your own to the instructor for written approval before beginning. You will use this exact domain for Assignments 2 and 3:

- Food delivery platform (orders, restaurants, menus, drivers, customers, real-time ETA)
- Fantasy sports league (players, teams, rosters, weekly scoring, trade history, standings)
- Campus IoT / smart building (buildings, rooms, sensor devices, time-series readings, maintenance tickets)
- Ride-share service (riders, drivers, vehicles, live trips, surge pricing, ratings)
- Music or podcast streaming (users, artists, albums, tracks, playlists, stream events, recommendations)
- Hospital or clinic scheduling (patients, providers, specialties, appointments, billing, prescriptions)
- Independent bookstore / e-commerce (inventory, categories, customers, orders, returns, reviews)

[UNIQUENESS] No two students should design an identical system. Even within the same domain, your choice of use cases, analytics goals, and AWS services should reflect your own thinking.

Assignment Requirements

Your slide deck must address all of the following sections. Each section maps to a graded rubric criterion.

1. Business Problem & Objectives (Slides 1-2)

- **Problem Statement:** Describe the business context and the core operational pain point your platform solves (2-4 sentences).
- **Success Criteria:** State 3-5 concrete, measurable objectives. Examples: reduce delivery late-rate by 15%; enable real-time player performance dashboards; detect anomalous sensor readings within 60 seconds.
- **Stakeholders:** Identify the primary stakeholders (e.g., operations managers, data analysts, ML engineers) and briefly describe what each one needs from this platform.

2. Solution Overview (Slide 3)

- **Platform Summary:** Summarize your platform in 3-5 bullet points. State: what data it collects, how it is processed and stored, and what analytical or ML outputs it produces.
- **Processing Mode:** Identify whether your solution is primarily batch, near-real-time, or real-time — and justify that choice for your domain.

3. Data Sources & Ingestion Requirements (Slide 4)

- **Source Inventory:** List all data sources your platform will consume. For each source specify:
 - Source name and type (e.g., OLTP database, IoT sensor stream, third-party REST API, manual batch upload)
 - Expected data format (JSON, CSV, Parquet, Avro, binary)
 - Approximate ingestion frequency (real-time, hourly, daily, weekly)
 - Estimated record volume per ingestion cycle
- **Internal vs. External:** Note which sources are internal (generated by your platform) vs. external (third-party APIs or partner data feeds).

4. AWS Architecture Diagram (Slides 5-6)

- **Architecture Diagram:** Create a labeled architecture diagram using draw.io, Lucidchart, or AWS Architecture Icons. The diagram must show all major components, data flows (arrows), and service boundaries.
- **Layered Architecture:** Organize the diagram using the following data-platform layers. Each layer must include at least one named AWS service:
 - **Layer 1 — Ingestion:** Ingestion Layer: how raw data enters your platform (e.g., AWS Kinesis Data Streams, AWS Glue, AWS DMS, S3 batch upload, API Gateway + Lambda).
 - **Layer 2 — Storage:** Storage Layer: where data is persisted at rest (e.g., Amazon S3, Amazon RDS, Amazon DynamoDB, Amazon Redshift, Amazon Neptune).
 - **Layer 3 — Processing:** Processing & Transformation Layer: how data is cleaned, enriched, and aggregated (e.g., AWS Glue ETL, AWS Lambda, Amazon EMR, Amazon Athena).
 - **Layer 4 — Analytics & ML:** Analytics & ML Layer: how insights and predictions are delivered (e.g., Amazon QuickSight, SageMaker, Athena, Redshift ML).
 - **Layer 5 — Security:** Security & Governance Layer: IAM roles, S3 bucket policies, encryption (KMS), VPC networking, and CloudTrail audit logging.

5. AWS Services Justification (Slide 7)

- **Per-Service Rationale:** For each AWS service in your architecture, provide a 1-2 sentence justification explaining why you chose it over a plausible alternative. Example: 'Amazon RDS MySQL was chosen over

Amazon DynamoDB because our domain requires complex multi-table JOIN queries for business reporting, which is well-supported by a relational engine.'

- **Trade-off Discussion:** Present at least one trade-off you considered: a service or design decision where you deliberated between two options and explain why you chose one over the other.

6. Data Sizing & Capacity Planning (Slide 8)

You must support all estimates with clearly stated business assumptions. Unfounded numbers will not receive full credit.

- **Record Volume Estimates:** Estimate total records/events generated per core data stream or activity per month (e.g., 'A ride-share platform at 10,000 active users completes ~50,000 trips per month, yielding ~50,000 trip records/month').
- **Storage Sizing Estimates:** Estimate average payload/record size in KB for each data stream/source and compute total monthly storage in GB.
- **Total Storage Growth (12 months):** Estimate S3 storage growth: raw landing bucket (Bronze), processed bucket (Silver), and analytics bucket (Gold) over a 12-month period.
- **Compute & Concurrency:** Estimate peak query concurrency (number of simultaneous analytical queries expected) and database connection requirements.

7. Cost Estimation & Business Value (Slides 9–10)

- **AWS Pricing Calculator:** Use the AWS Pricing Calculator (calculator.aws) to produce an itemized monthly cost estimate for all services in your architecture. Include a screenshot of the calculator summary and attach the shareable link.
- **Cost Breakdown Table:** Present the cost breakdown in a slide table with columns: Service, Monthly Cost (USD), Key Cost Driver (what primarily drives the bill — storage GB, compute hours, data transfer, API calls).
- **Total Estimate & Optimization:** Estimate total monthly cost across all services and discuss one concrete cost-reduction technique (e.g., reserved instances, S3 Intelligent-Tiering, query result caching, right-sizing).
- **Business Value Statement:** In 2–3 sentences, quantify the business value this platform enables. Where possible, use metrics (e.g., 'Reducing late deliveries by 15% on 10,000 daily orders at \$5 average order value translates to \$27,500/month in improved customer retention revenue').

8. Appendix (Optional Additional Slides)

- Alternative architecture options that were considered and rejected (with reasoning).
- Detailed AWS Pricing Calculator screenshots or line-item tables.
- Data governance, compliance, or regulatory notes (HIPAA, GDPR, CCPA) relevant to your domain.
- Disaster recovery / multi-region considerations.

Slide Deck Requirements

Your slide deck must meet the following structural and quality standards:

- **Length:** 10–15 slides total (excluding title slide and appendix). Every slide must have a clear, descriptive title.
- **Visual Communication:** Use diagrams, tables, and bullet points — avoid walls of paragraph text on slides.
- **Consistency:** Every AWS service shown in the architecture diagram must also appear in the services justification table.
- **Format:** Submit as a PDF export of your slide deck. Optionally include the editable source (.pptx or Google Slides link) in the appendix.
- **Recommended Slide Flow:** Use the following suggested slide order:

Slide 1: Title slide (course, assignment number, your name, domain chosen)

Slide 2: Business Problem & Objectives

Slide 3: Solution Overview (platform summary, batch vs. streaming)

Slide 4: Data Sources & Ingestion Requirements

Slide 5–6: AWS Architecture Diagram (organized by data platform layers)

Slide 7: AWS Services Justification & Trade-offs Table

Slide 8: Data Sizing & Capacity Planning

Slide 9: Cost Estimation Table (from AWS Pricing Calculator)

Slide 10: Business Value & Cost Optimization

Slide 11+: Appendix slides as needed

Grading Rubric

Total: 100 points

Criterion (Points)	Exceeds Expectations	Meets Expectations	Below Expectations
Business Problem & Objectives (10)	Clear, specific problem statement with measurable success criteria and well-defined stakeholder roles.	Problem and objectives stated; success criteria present but may lack precision.	Vague problem description, missing success criteria, or no stakeholder identification.
Data Sources & Ingestion Specifications (15)	All sources comprehensively inventoried with format, frequency, volume estimates, payload characteristics, and clear distinction of internal vs. external data feeds.	Data sources listed with most required details (format, frequency, approximate volume).	Incomplete source list, missing format or volume details, or no distinction of internal/external sources.
AWS Architecture Diagram (25)	All 5 layers present; every service labeled and justified; data flows (arrows) clearly shown; diagram is clean, professional, and follows Well-Architected principles.	Architecture diagram covers all 5 layers with named AWS services and data flows.	Diagram missing one or more layers, uses unnamed generic boxes, or data flows are absent.

Criterion (Points)	Exceeds Expectations	Meets Expectations	Below Expectations
Services Justification & Trade-offs (20)	Each service justified with specific technical reasoning; at least one thoughtful architectural trade-off analyzed with both alternatives thoroughly evaluated.	Most services justified; at least one trade-off discussed.	Justifications are generic (e.g., 'S3 is good for storage'), or no trade-off discussion.
Capacity Planning & Storage Sizing (15)	Record counts, storage per data stream, and 12-month growth across bronze/silver/gold tiers all estimated with clearly stated, realistic business assumptions.	Record counts and storage estimates provided; assumptions stated, though some may be rough.	Estimates missing, not grounded in stated assumptions, or clearly unrealistic.
Cost Estimation & Business Value (15)	Itemized cost table from AWS Pricing Calculator with shareable link; optimization technique identified; quantified business value.	AWS Pricing Calculator used; cost breakdown table present; brief business value statement.	Cost estimate not from AWS Calculator, missing breakdown, or no business value discussion.

Submission Deliverables

Deliverable	Format	Notes
Slide Deck	PDF (required)	10–15 slides covering all required sections. Export from PowerPoint, Keynote, or Google Slides as PDF.
AWS Pricing Calculator Link	URL in the slide or report	Shareable link from calculator.aws showing your itemized estimate.
Architecture Diagram Source	Optional (.drawio, .pptx, or Google Slides link)	Include in appendix or submit separately. Required only if the embedded diagram is not legible in PDF.

[SECURITY REMINDER] Do not include AWS account credentials, IAM access keys, or any real login details in your slides or appendix.